East West University

Department of Mathematical and Physical Sciences

Semester: Fall 2025

Course Outline: MAT 104

Course Title : **Co-ordinate Geometry and Vector Analysis**
Course Code : MAT 104
Pre-Requisite : MAT 101
Credit Hour : 3
Course Instructor : Marzia Yesmin(MY), Dept. of MPS.
Office Room : Room No.: FUB 1004.
E-mail Address : marzia.yesmin@ewubd.edu
Contact No. : +8801616 297979
Class Start : 28/09/2025
Class End : 18/12/2025

Class Hours:

Section	Days	Time	Room No.
14	Sunday & Thursday	11.50 am – 1.20 pm	FUB 602

Office Hours:

Sunday	Thursday	Room No.
10.50 am – 11.50 am	10.50 am – 11.50 am	FUB 1004

Course Goal:
This course is designed to develop the basic concepts of students in the area of coordinate geometry and vector analysis. A student has to study this course because the contents of the course are very applicable in all branches of Science and Engineering. Also, the methods of vector analysis provide a natural aid to the understanding of geometry and some physical concepts. They are also a fundamental tool in many theories of Applied Mathematics. Our goal is to obtain basic knowledge about two-dimensional geometry, three-dimensional geometry, and vector analysis

Course Learning Outcomes:
By the end of this course, students should be able to:
(a) identify the coordinates of a point (b) transform the coordinates of a point or the equation of a curve (c) identify the curves (d) calculate scalar and vector products (e) find the vector equations of lines and planes (f) differentiate vector functions of a single variable (g) calculate velocity and acceleration vectors for moving particles (h) find the gradient of a function. (i) find the divergence and curl of a vector field (j) use

the gradient operator to calculate the directional derivative of a function (k) calculate the unit normal at a point on a surface (l) recognize irrotational and solenoidal vector fields (m) evaluate line and surface integrals (n) understand the various integral theorems relating line, surface and volume integrals.

Course Contents: 1. Two-Dimensional Geometry
2. Three-Dimensional Geometry
3. Vector Analysis

Description: The following topics will be covered throughout the semester.

Two-Dimensional Geometry

Change of axes. Transformation of co-ordinates. Pair of straight lines. Circles: Tangents and Normals, Chord of contact, System of circles, Orthogonal circles. Conic Section: Parabola, Ellipse and Hyperbola. The general equation of the second degree, Identification of curves.

Three-Dimensional Geometry

Coordinate system. Direction cosines and direction ratios. Plane, Straight line.The shortest distance. Sphere: Tangent plane. Cylinder and Cone.

Vector Analysis

Vectors and Scalars. Algebra of vectors, Vector differentiation and vector integration. Gradient. Divergence and Curl. Cartesian, Spherical, Polar and Cylindrical systems. Physical significance of Gradient, Divergence and Curl. Green's Theorem, Divergence Theorem. Stoke's Theorem and its applications.

Text Books: **(a)** A Text Book on Co-ordinate Geometry and Vector Analysis: Khosh Mohammad **(b)** Vector Analysis: Schaum's Outline Series.

Reference Books: **(a)** A Text Book on Co-ordinate Geometry with Vector Analysis: Rahman & Bhattacharjee.
(b) Co-ordinate Geometry: Luther Pfahler, Dover Publications Inc.

Term Examinations:

Examinations	Dates
Quiz-I	19th October, 2025 (Sunday)
Quiz-II	6th November, 2025 (Thursday)
Mid Term	16th November, 2025 (Sunday)
Quiz-III	7th december, 2025 (Sunday)
Assignment & Presentation	14th December, 2025 (Sunday)
Final	22nd December, 2025 (Monday)

Score Distribution:

Assessment Tools	Marks
1. Final	30
2. Mid Term	25
3. Quiz	20
3. Assignment	10
4. Presentation	05
5. Class Attendance	05
6. Class Performance	05

Grading Policy:

Numerical Scores	Letter Grade	Grade Point
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
J S% to less than 50%	C	2.25
JO% to less than 45%	D	2.00
Less than 40%	F	0.00

Special Guidelines:

♣ There will be 3 quizzes taken during the semester. The best 2 will be counted for each term.
♣ Students are strongly recommended not to miss any quizzes. No make-up quizzes will be held.
♣ Students are requested to silence their mobile phones during the class hour.
♣There is zero tolerance for cheating at EWU. Students caught with cheat sheets in their possession, whether used or not used, &/or copying from cheat sheets, writings on the palm of a hand, back of calculators, chairs, or nearby walls, etc., would be treated as cheating in the exam hall. The only penalty for cheating is expulsion from EWU.

Marzia Yesmin

Date: 26/09/2025